GlucoTwin System Architecture

Multimodal Predictive Digital Twin for Proactive Glucose Risk Surveillance & Counterfactual Exploration

1. MULTIMODAL SYNTHETIC DATA

Simulated Clinical & Sensor Ingestion

Historical Synthetic EHR

[Verified Stored Context]

Demographics & Phenotype
Age, Subtype, Weight, Height, BMI

Historical Diagnoses
HTN, Neuropathy, Dyslipidemia

Structured Lab Results
HbA1c, Fasting Glucose, Lipids

Genetic Markers (EHR)
TCF7L2, PPARG, KCNJ11 Risk Alleles

Dynamic Physiological Streams

[15-Min Sampling Schedule]

Continuous Glucose (CGM)
15-min intervals (40-400 mg/dL)

Cardiac Autonomic Vitals
Heart Rate (bpm) & HRV (RMSSD)

Sleep Architecture Stream
Deep, Light, REM Stages & Duration

Physical Activity Telemetry
15-min Interval & Daily Cumulative Steps

Synthetic EHR & Labs

CGM, Vitals, Sleep, Steps

2. BACKEND & TWIN INTELLIGENCE

FastAPI Core (Render Deployment: glucotwin-backend.onrender.com)

Data Ingestion & Controlled Simulation Feed

Pydantic Schemas & Bounded Idempotence

Simulated Sensor Epoch Feed
POST /api/digital-twin/simulate-feed

Idempotent Deduplication
SQLite Unique Index on Alerts & Records

Validated Records

Digital Twin State Vector & Feature Engineering

GET /api/digital-twin/{patient_id}

Twin Dynamic Aggregation Engine
Calculates 24h Baselines vs 1h Deltas (ΔG, ΔHR, ΔHRV), Sleep & Activity Summaries

14-Feature ML Vector (Leakage-Purged)
Current G, 1h Mean, 1h StdDev, Trend Slope, HR, HRV, Circadian sin/cos, Subtypes

14-Feature Input

ML Glucose Spike Engine

Logistic Regression + Scaler

Binary Classifier (y >= 180)
120-min Forward Horizon
Threshold >= 180 mg/dL
ROC-AUC: 0.825
Strict Purged Holdout

Baseline Telemetry

Alerts & Counterfactual

Surveillance & What-If

Dual-Source Alert Engine
Observed (<54, >250) vs Model

What-If Sandbox (Heuristic)
ΔG = Carbs*(0.40*K) - Act*(0.75)
Illustrative Educational Sandbox

Patient API

Trigger +15m Sensor Ingestion

Digital Twin API

Spike Pred API

Alerts / What-If API

3. CLINICIAN DASHBOARD & TRIAGE

React 19 / Vite SPA (Vercel: gluco-twin-eta.vercel.app)

Patient Cohort Selector
5 Diverse Clinical Personas
Interactive Subject Switcher

Multimodal Historical EHR Panel
Longitudinal HbA1c, Fasting Glucose, Lipids,
5 Genetic Marker Alleles with Synthetic Labels

Dynamic Sensor Telemetry
Synchronized CGM, HR, HRV Curves,
Sleep Hypnogram Stages & 15-Min Step Bars

Digital Twin & Live Simulation
State Vector Deltas, Sensor Coverage,
+15m Simulated Sensor Feed Workflow

Predictive Risk & What-If
2h Spike Forecast with Explanations,
Counterfactual Meal/Activity Sliders

Alerts & Clinician Triage Pager
Active, Acknowledged, Resolved Feed
Simulated Hospital Pager Dispatch
[Explicitly Disclaimed: Local Demo Only,
No External SMS/Email Sent]

LEGEND: Solid Blue Lines = Implemented Automated Data Pipeline | Dashed Red Line = Demonstrable Interactive Simulated Sensor Update Workflow | Purple = Digital Twin Processing State | Green = Validated ML Model

REGULATORY & ETHICAL NOTICE: GlucoTwin is an academic engineering research proof-of-concept for the Happiest Health Digital Twin Challenge. All telemetry, laboratory results, genetic markers, and clinician pagers are synthetic. Not for medical diagnosis, treatment, or clinical intervention.